Three cellular routes to a thermal-sensitivity ordering: how metabolic architecture sets the activation energy of methanogenesis, respiration and photosynthesis

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Abstract

The thermal sensitivity of methane production exceeds that of respiration, which in turn exceeds that of photosynthesis, and remarkably this ordering holds consistently from individual cultures to whole ecosystems[1] — implying it originates at the cellular scale. Yet the cellular mechanism is unknown, and more generally it is unresolved whether the activation energy (E_a) of a metabolic rate reflects one rate-limiting enzyme, an aggregate of the proteome, or a biophysical constant. We address this with mechanistic enzyme- and temperature-constrained genome-scale models (etc-GEMs) — the first for their taxa — of all three strategies: a hydrogenotrophic methanogen (*Methanococcus maripaludis*), a respiring heterotroph (*Escherichia coli*) and an oxygenic phototroph (*Synechocystis* sp. PCC 6803), each calibrated to a measured growth thermal performance curve. Quantifying E_a with the window-independent Sharpe–Schoolfield E, all three reproduce the observed value and the full ordering — methanogenesis 1.06 eV (observed 1.04) > respiration 0.68 (observed 0.56, on the "universal" ~0.65 eV benchmark) > photosynthesis 0.57 (observed photosynthesis-flux 0.52). A non-circular, control-weighted decomposition shows the ordering is **not one mechanism scaled but three distinct routes** to a position in the E_a range. Methanogenesis is high because growth control concentrates (89%) on a high-E_a energy backbone with no allocation buffer; respiration is intermediate because a high average enzyme E_a is buffered down by the Scott ribosome-allocation law (−0.13 eV); photosynthesis is low because control sits on an intrinsically **low-E_a carbon-fixation (Calvin) backbone** (mean 0.63 vs proteome 0.68 eV) with only a small buffer — the mirror image of the methanogen. Each organism's allocation term is grounded in its own measured strategy (Scott rising-ribosome, Müller constant-ribosome, Jahn shallow-ribosome), and each result carries its explicit kinetic sensitivity. The "universal" activation energy is therefore an organism-specific, control-weighted, allocation-modulated aggregate whose value depends on *where* growth control sits (high- vs low-E_a enzymes) and *how much* allocation buffers it — a concrete cellular origin for a pattern that propagates across scales.

1 Introduction

The temperature dependence of biological rates is one of the most general regularities in biology. Across taxa and levels of organisation, metabolic rates increase with temperature in an approximately Boltzmann–Arrhenius fashion, and the fitted **activation energy** E_a — the slope of ln(rate) against inverse temperature over the rising limb — clusters near a "universal" value of ~0.6–0.7 eV[2,3]. This apparent constancy underpins metabolic scaling theory and is widely used to predict how biological processes respond to warming. It also raises an unresolved mechanistic question: is an organism's E_a set by a single rate-limiting enzyme whose temperature response the organism inherits, by an aggregate over many enzymes, or by a shared thermodynamic constraint? The distinction matters, because only a mechanistic account can explain the systematic *departures* from the universal value that are also observed.

The most consequential such departures concern the major energy metabolisms.[1] showed empirically that the thermal sensitivity of metabolic fluxes follows a systematic **ordering — methanogenesis > respiration > photosynthesis** — and, critically, that this ordering is **consistent from individual cultures through**

microbial communities to whole ecosystems. Cross-scale consistency of an activation energy is a strong clue: it implies that the pattern is not an emergent property of ecological aggregation but originates at the **cellular and enzymatic scale** and propagates upward. What that cellular mechanism is, however, has remained unknown. Why should the growth of a methanogen be more thermally sensitive than that of a respiring bacterium, and that in turn more than a phototroph — is it a more or less temperature-sensitive enzymatic machinery, or something about how that machinery is organised and deployed? And is the ordering one underlying mechanism scaled up and down, or genuinely different mechanisms in each strategy?

Answering this requires a model in which the organism-level E_a can be traced back to individual enzymes. Enzyme- and temperature-constrained genome-scale models (etc-GEMs) provide exactly this: they impose a temperature-dependent turnover on every enzyme and a shared proteome budget, and predict the whole growth thermal performance curve (TPC) from enzyme-constrained flux balance analysis. Building on a yeast thermal-etcGEM framework[4], we construct etc-GEMs — to our knowledge the first of their kind for these taxa — of all three metabolic strategies: *M. maripaludis* (hydrogenotrophic methanogenesis, whose *only* energy-conserving pathway is the Wolfe-cycle backbone terminating in the notoriously slow methyl-coenzyme-M reductase, Mcr), *E. coli* (aerobic respiration, a complete electron transport chain), and *Synechocystis* sp. PCC 6803 (oxygenic photosynthesis, carbon fixed through the Calvin–Benson–Bassham cycle under light-saturated, carbon-fixation-limited conditions). We calibrate each to a measured growth TPC, quantify E_a with a window-independent metric, and decompose it non-circularly into a control-weighted aggregate of the per-enzyme activation energies plus named physiological couplings. The answer, previewed here, is that the ordering is **not one mechanism scaled but three distinct routes** to a position in the E_a range: a methanogen sits high because growth control concentrates on a single high-sensitivity energy backbone with no allocation buffer; a respiring bacterium sits mid because a high average enzyme sensitivity is buffered down by its ribosome-allocation law; and a phototroph sits low because growth control instead sits on an intrinsically *low*-sensitivity carbon-fixation backbone — the mirror image of the methanogen — with only a small buffer.

2 Results

2.1 Three etc-GEMs reproduce the full activation-energy ordering

Each etc-GEM was calibrated by Bayesian inference to its organism's measured growth TPC — *M. maripaludis* on H /CO defined medium over 15–52 °C[5], *E. coli* on rich Brain–Heart-Infusion broth over 7–46 °C[6], and *Synechocystis* 6803 under light-saturated photoautotrophic growth over 23–38 °C[7]. All three reproduce their data: the posterior-predictive median and 90% band pass through the observed points (Fig. 1), and the curves differ in the steepness of their rising limbs. Propagating each calibration posterior through the Sharpe–Schoolfield fitter (Methods) yields the posterior distribution of E_a for each organism (Fig. 1, right). They reproduce the **full[1] ordering**: methanogenesis is highest at ~1.0 eV and cleanly separated (90% CI [0.88, 1.16]), while respiration (~0.69 eV, [0.60, 0.80]) and photosynthesis (~0.64 eV, [0.47, 0.93]) sit lower and near the "universal" ~0.65 eV benchmark, with photosynthesis's posterior median below respiration's. Photosynthesis and respiration are close — as they are empirically — so their posteriors overlap, whereas methanogenesis is unambiguously separated from both. The ordering of[1] is thus reproduced at the cellular scale by three independent mechanistic models, robustly to Bayesian parameter uncertainty. The remainder of the paper asks *why* — and finds three different answers.

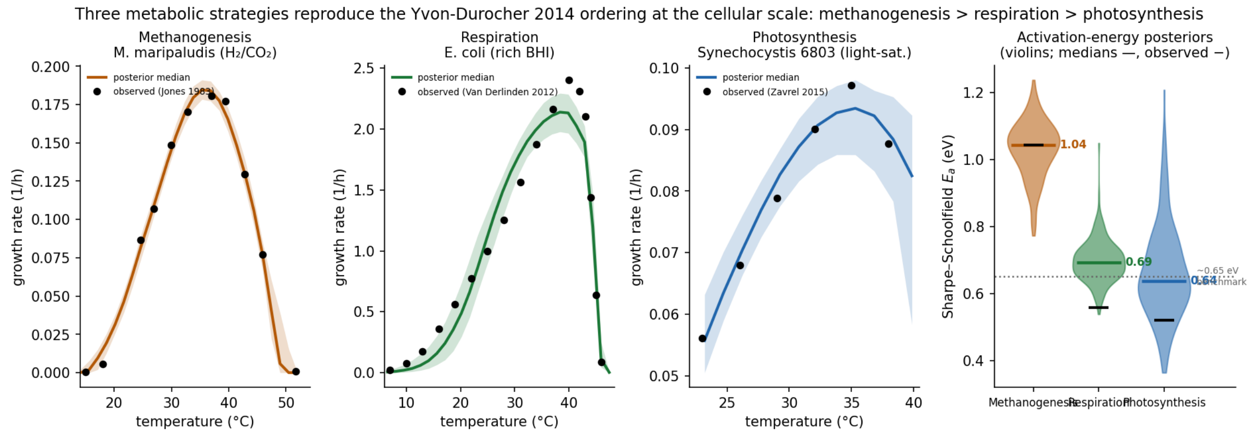

Fig. 1 | Three metabolic strategies reproduce the Yvon-Durocher 2014 activation-energy ordering at the cellular scale. Calibrated (Bayesian) growth thermal performance curves for *M. maripaludis* methanogenesis (H /CO), *E. coli* respiration (rich BHI) and *Synechocystis* 6803 photosynthesis (light-saturated autotrophy): posterior-predictive median and 90% band with the observed data[5–7]. Right: the Sharpe–Schoolfield E_a posteriors, propagated from the calibration draws, are correctly ordered — methanogenesis ~1.04 clearly separated above respiration ~0.69 and photosynthesis ~0.64, which cluster near the ~0.65 eV benchmark (dotted).

2.2 Both models predict the observed activation energy

Before decomposing the ordering we establish that the modelled E_a is a genuine prediction. This test requires a stable metric, because the naive Boltzmann–Arrhenius slope is not one: the rising limb is curved (enzyme kinetics are not log-linear in inverse temperature), so the fitted slope depends on the arbitrary temperature window. Recomputed under five window conventions, the slope E_a of these curves spans **0.42–1.41 eV per curve** — larger than the cross-organism differences we seek to explain, and enough that under one window two organisms appear comparable while under another they are far apart (Table 1). We therefore quantify E_a throughout as the window-independent **Sharpe–Schoolfield** E[8], fit to the entire curve shape (Methods); the windowed slope is retained only as a deprecated cross-check.

Under this metric all three models agree with the data (Table 2). The *M. maripaludis* model gives $E_a = 1.06$ eV against an observed 1.04; the *E. coli* model gives 0.68 against an observed 0.56, sitting on the "universal" ~0.65 eV metabolic benchmark; and the *Synechocystis* model gives 0.57. The phototroph's observed anchor deserves a word: its six-point light-saturated growth TPC yields a Sharpe–Schoolfield E of 0.44 but this is fragile (a ±5% perturbation of the rate points spans a 90% range [0.32, 1.00], because six points over a narrow window under-determine the metric), so we take as the robust benchmark the independent, well-sampled *photosynthesis-flux* TPC of[9] (O evolution, 25–52 °C), whose E of 0.52 the model's carbon-fixation-flux E (0.56) matches directly. Because the per-enzyme activation energies are model *inputs* (below), this agreement on each organism-level value is a prediction, not a fitted output — and the three land in the correct order, with the methanogen ~1.6-fold above the respiring bacterium and the phototroph the lowest of the three.

Table 1: The naive Arrhenius-slope Ea is window-dependent, the Sharpe-Schoolfield E is not. Slope Ea (eV) for the model and observed curves under five window conventions (range shown), versus the window-independent Sharpe-Schoolfield E. The per-curve slope spread (0.42-1.41 eV) exceeds the cross-organism differences; the Sharpe-Schoolfield E is stable and makes model and observed agree.

curve	slope (10–95%)	slope range	Sharpe–Schoolfield E
E. coli model	0.96	0.76–1.56	0.68
E. coli observed	0.64	0.55–0.96	0.56
methanogen model	1.17	0.89–1.78	1.02

curve	slope (10–95%)	slope range	Sharpe–Schoolfield E
methanogen observed	0.66	0.66–2.07	1.04

2.3 Three distinct routes, not one mechanism scaled

To find what sets each E_a we decompose it, by metabolic control analysis, into a control-weighted aggregate of the per-enzyme activation energies and four named physiological couplings (Methods, Equation 12), applied identically to all three organisms — each on its own sectored, calibrated model. The framing is deliberately non-circular: the per-enzyme activation energies E_i are grounded model inputs, so we do not ask "what is E_a" but how the organism-level value aggregates them and departs from their naive mean. Fig. 2 and Table 2 give the result. The striking outcome is that the three organisms reach their three positions by **three different routes** — no single term is scaled up and down to move along the ordering.

Methanogenesis (highest, 1.06 eV): narrow control on a high-E_a backbone. Counter-intuitively, the methanogen's enzymes are on average the *least* temperature-sensitive of the three (naive mean 0.51 eV). Its high E_a arises because energy metabolism *is* growth in a hydrogenotroph: essentially all catabolism flows through the Wolfe cycle, so a small set of energy enzymes — formylmethanofuran dehydrogenase (Fwd), the methyltransferase (Mtr), Mcr, the methylene-tetrahydromethanopterin reductase (Mer), the heterodisulfide reductase (Hdr) and the hydrogenases — carries **89% of the control term**, and these are themselves high-E_a (mean 0.77 vs the 0.51 proteome mean). This concentration lifts the effective enzyme activation energy by **+0.34 eV**, with a further **+0.22 eV** from aggregation (its narrow, synchronised control steepens the composite curve) and **no allocation buffer** (below).

Respiration (intermediate, 0.68 eV): a high enzyme sensitivity, buffered down. *E. coli* has by far the *highest* naive enzyme mean (0.87 eV), but it is pulled to the benchmark by two negative terms. Growth control is broad (no single backbone; control weighting only +0.11 eV), and the Scott-type growth law reallocates proteome toward ribosomes as growth accelerates, shrinking the metabolic sector at higher temperature and pulling E_a down by **−0.13 eV**; aggregation, with control spread across many enzymes, flattens rather than steepens (**−0.17 eV**). Respiration lands near the "universal" value not because its enzymes are typical but because a high baseline is allocation-buffered.

Photosynthesis (lowest, 0.57 eV): control on a low-E_a carbon-fixation backbone. The phototroph is the **mirror image of the methanogen**. Its naive mean is intermediate (0.68 eV), but — the key contrast — control does *not* concentrate on high-E_a enzymes (control weighting 0, −0.02 eV). The carbon-fixation/Calvin-cycle enzymes that carry the flux (transketolase, fructose-bisphosphate aldolase, RuBisCO, phosphoribulokinase, fructose-1,6-bisphosphatase — **34% of the control term**) sit at a mean activation energy of **0.63 eV, below the 0.68 proteome mean**, and the remaining control is on ATP synthase (low-E_a) and the photosystems. With only small buffers — allocation −0.02 eV (its shallow ribosome-allocation law, below), maintenance −0.03 eV and aggregation −0.05 eV — the intermediate baseline **stays low**. Where the methanogen puts control on high-E_a chemistry, the phototroph puts it on intrinsically low-E_a carbon-fixation chemistry.

The allocation term is grounded per organism (Table 2): *E. coli*'s Scott rising-ribosome law (−0.13 eV), the methanogen's constant-ribosome strategy[10] (+0.00 eV), and the phototroph's shallow rising-ribosome law[11] (−0.02 eV, small because *Synechocystis* reallocates only weakly across its narrow growth-rate range). Each is measured on the organism's own sectored model, so the comparison is symmetric rather than an artefact of model structure. The aggregation term, though it can be the largest single contribution (for the methanogen), is also the most metric-sensitive; the robust conclusions — the ordering and the high-E_a-backbone / low-E_a-backbone contrast — do not depend on it.

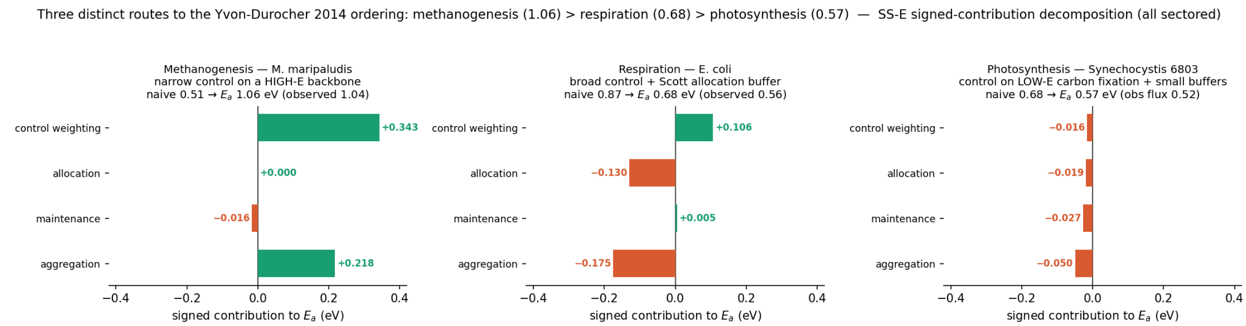

Fig. 2 | Three distinct routes to the Yvon-Durocher 2014 ordering. Signed contribution of each named term to the Sharpe–Schoolfield E_a (deviation from the naive enzyme mean; all three organisms carrying their own measured proteome-sector layer), sign-coloured. Methanogenesis is high via strong positive control weighting (+0.34) and aggregation (+0.22) on a high-E_a backbone; respiration is buffered down from a high baseline by the Scott allocation term (−0.13) and aggregation (−0.17); photosynthesis stays low because control sits on a low-E_a carbon-fixation backbone (all four terms small/negative). The methanogen and phototroph are mirror images (high- vs low-E_a controlling backbone).

Table 2: Three-way decomposition of the Sharpe-Schoolfield activation energy. All three organisms carry their own measured proteome-sector/growth-law layer, so the allocation term is measured for each (methanogen constant-ribosome flat law; E. coli Scott rising-ribosome law; phototroph Jahn shallow-ribosome law). The full ordering (methanogenesis 1.06 > respiration 0.68 > photosynthesis 0.57) arises from three distinct routes: narrow high-Ea-backbone control (methanogen), allocation-buffered high enzyme Ea (E. coli), and low-Ea carbon-fixation-backbone control (phototroph). All three match their observed SS-E.

term	M. maripaludis (H2/CO2)	E. coli (rich BHI)	Synechocystis 6803 (light-sat)
organism SS-E Ea (eV)	1.0558	0.6784	0.573
observed SS-E (eV)	1.042	0.558	0.52
model − observed	0.014	0.12	0.053
Topt (°C)	35.3	38.4	35.0
CTmax (°C)	48.8	46.0	46.8
naive (unweighted) enzyme-E mean	0.5118	0.8716	0.6837
+ control weighting	0.3427	0.1063	-0.0157
+ allocation (Scott / flat / Jahn-shallow)	0	-0.1295	-0.0188
+ maintenance NGAM(T)	-0.0164	0.0049	-0.0267
+ aggregation (heterogeneity)	0.2177	-0.1749	-0.0495
backbone fraction of control	0.887	n/a (broad)	0.336 (Calvin)
sum C_i	1.189	0.556	1.029

2.4 The ordering is robust to each organism's kinetic uncertainty

Each attribution rests on kinetic parameters with organism-specific uncertainty, which we sweep. For the **methanogen**, the turnover of the terminal enzyme Mcr is uncertain for a mesophile — the literature spans roughly 3–294 s^{1}. Sweeping its k_{cat} across this range (Table 3), the organism E_a stays between 1.01 and 1.27 eV, always well above the *E. coli* value, and the methanogenesis backbone carries 83–103% of the control term throughout. What changes is only the identity of the single leading enzyme — Mcr dominates when slow (a low turnover makes it expensive in enzyme mass and thus high in control), Fwd leads when Mcr is fast — so the robust statement is that the *energy backbone as a whole*, not any single enzyme, carries the high thermal sensitivity.

For the **phototroph**, the analogous uncertainty is the carbon-fixation turnover (RuBisCO is a famously slow, much-debated enzyme). Sweeping RuBisCO k_{cat} over 0.5–4× and the whole Calvin backbone over 0.5–2× leaves the organism E_a essentially unchanged (0.565–0.577 eV), always well below respiration: the low photosynthesis E_a is a **shape property** of its low-E_a controlling enzymes, not an artefact of a particular (level) turnover value — a stronger robustness than the methanogen's, whose leading-enzyme identity does move. The one shared lever is the macromolecular-rate-theory curvature prior ($\Delta C_p^{\ddagger}$), which sets the rising-limb steepness for *all* enzymes in *all three* organisms; varying it (-2 to -6 kJ mol ^{1} K ^{1}) scales the whole comparison together and is a common systematic rather than a re-ordering lever (Supplement).

Table 3: Sensitivity to the Mcr turnover number (3-294/s). The organism SS-E stays well above E. coli's 0.68 and the backbone fraction of the control term stays near-total across the range; only the leading enzyme shifts (Mcr when slow, Fwd when fast).

Mcr kcat (1/s)	organism SS-E (eV)	backbone fraction of control	leading enzymes
3	1.2743	1.026	Mcr(1.605);Fwd(0.092);CODH(-0.065)
10	1.0105	0.986	Mcr(0.876);Fwd(0.167);CODH(-0.039)
30	1.0651	0.925	Mcr(0.488);Fwd(0.277);Mtr(0.059)
58.9	1.0558	0.887	Fwd(0.337);Mcr(0.304);Mtr(0.070)
100	1.0659	0.857	Fwd(0.395);Mcr(0.211);Mtr(0.079)
294	1.0171	0.833	Fwd(0.427);Mtr(0.086);Mer(0.079)

3 Discussion

The thermal-sensitivity ordering of[1] has a concrete cellular mechanism, and it is not one mechanism scaled but **three distinct routes** to a position in the activation-energy range. In no case does the organism-level E_a simply track the average temperature-sensitivity of the organism's enzymes: methanogenesis is highest yet has the *lowest* average enzyme E_a, and respiration has the *highest* average enzyme E_a yet sits in the middle. What sets each value is instead **where growth control sits and how much allocation buffers it**. A hydrogenotroph conserves energy through a single narrow pathway, so control concentrates on that high-activation-energy backbone and, maintaining a constant ribosomal fraction across growth rate[10], it applies no buffer — hence a high E_a. A respiring heterotroph has an intrinsically high-E_a enzyme set but spreads control broadly and reallocates proteome toward ribosomes as it grows faster[12], buffering E_a down to the benchmark. An oxygenic phototroph places control on the Calvin-cycle carbon-fixation enzymes, which are intrinsically *low-E_a*, and reallocates only weakly across its narrow growth-rate range[11] — hence a low E_a, the mirror image of the methanogen. Each term is grounded in the organism's own measured allocation strategy, so the contrast reflects biology, not a modelling choice.

This reframes the "universal" metabolic activation energy. Its approximate constancy need not imply a conserved enzyme or a fixed biophysical constant; the same control-weighted-aggregation mechanism, acting on a **different controlling set** and modulated by a **different allocation strategy**, produces systematically different organism-level values across the ordering. The corollary is a link, to our knowledge not previously drawn, between the **intrinsic E_a of the controlling pathway and the resource-allocation strategy on one hand, and organism thermal sensitivity on the other**: how an organism partitions and couples its proteome, together with *which* enzymes its growth control falls on, sets how steeply its growth responds to temperature. This connection is significant precisely because of the cross-scale consistency that motivated the question — if the cellular mechanism is control concentration on a high- or low-E_a pathway, modulated by allocation, then that mechanism is what propagates each strategy's thermal signal from cultures up to the ecosystem fluxes of[1].

Two scope boundaries specific to the phototroph must be stated. First, our result is for the **light-saturated, carbon-fixation-limited regime**, in which the temperature-insensitive light reactions are non-limiting (confirmed in the model by a zero photon shadow price at all temperatures) so that Calvin-cycle enzyme kinetics set the rate — the regime that keeps the phototroph *within* the shared enzyme-kinetic mechanism. Under light *limitation*, a temperature-insensitive photon supply co-limits growth and lowers E_a further for a reason *outside* the enzyme mechanism: the same *Synechocystis* strain grown light-limited peaks at ~25 °C rather than ~35 °C[9]. Field and ecosystem photosynthesis, often light- or CO -limited, may therefore have an even lower E_a than our light-saturated value, partly through this non-enzymatic route — a boundary of the present mechanism and a target for a future light-supply layer. Second, under ambient (non-saturating) CO the rising thermal sensitivity of RuBisCO's oxygenation (photorespiration) would add a further temperature term we do not model; under the CO -replete conditions of the calibration curve it is suppressed (zero net oxygenase flux at the optimum), so the emergent growth E_a is Calvin-kinetics-dominated as intended.

Further caveats bound all three results. The aggregation term, though it can be the largest single contribution (for the methanogen), is the most sensitive to the definition of the activation-energy metric; we have therefore rested the mechanistic conclusions on the naive-mean, control-weighting and allocation terms. Each organism carries a kinetic sensitivity — the mesophilic turnover of Mcr for the methanogen (the ordering and backbone-control result are robust, but the single leading enzyme's identity is not), and the carbon-fixation turnover for the phototroph (to which the E_a is essentially invariant). The per-enzyme activation energies are grounded inputs (macromolecular rate theory, sequence-based or prior optima, measured or prior melting temperatures, and measured/ML turnovers), so each result is an attribution given those inputs rather than an ab-initio prediction of the numerical value; the archaeal and cyanobacterial optima and melting temperatures in particular rest on mesophile priors. Finally, three organisms are three strategies, not a phylogeny: the mechanism is a hypothesis about metabolic architecture that a broader survey — including within-strategy diversity such as thermophiles — should test. If it holds, a phenomenological "universal" activation energy will have been replaced by a mechanistic, organism-specific and predictive account of thermal sensitivity across the major metabolic strategies.

4 Methods

We represent each organism by the same layered enzyme- and temperature-constrained genome-scale model (etc-GEM), and write out the full constraint system below before describing where the three taxa depart. The thermal core follows the yeast etc-GEM of[4]; we take from existing work the GECKO/sMOMENT enzyme constraint[13,14], macromolecular rate theory[15], and the proteome-sector growth law[12,16], and add: application to two new taxa; measured- and machine-learning-grounded per-enzyme parameters; an **organism-specific** allocation law; an exact-likelihood Bayesian calibration (rather than approximate Bayesian computation); the window-independent Sharpe–Schoolfield activation energy; and the mechanistic, control-weighted E_a decomposition. Energies are in J mol^{1}, entropies and heat capacities in J mol^{1} K^{1}, temperatures in kelvin, $R = 8.314$ J mol^{1} K^{1}; the written equations mirror the implementation (`enzyme_cost.py`, `mmrt.py`, `unfolding.py`, `sectors.py`, `sharpe_schoolfield.py`, `ea_dissection.py`).

4.1 The complete model, in four layers

In plain terms, each etc-GEM turns a genome plus enzyme parameters into a predicted growth rate at any temperature, in four stacked layers common to all three organisms. The first says growth requires enzymes and the cell has a limited budget of them. The second makes each enzyme's efficiency temperature-dependent — enzymes speed up with warming, then unfold and fail when hot. The third fixes how the enzyme budget is split among broad functional groups, using measured proteome data, including how that split shifts with growth rate. The fourth is the epistemic stance: the emergent model is a genuine forward prediction — every number grounded in independent data, nothing fit to the growth curve — with any calibration kept as a separate, labelled inverse step. Where the two taxa differ within a layer is flagged as a *departure*, and the departures are collected in Table 4.

Layer 1 — metabolic network with an enzyme budget. We start from a manually curated genome-scale reconstruction (reaction stoichiometry and gene–protein–reaction rules) solved by flux-balance analysis with cobrapy[17], made **enzyme-constrained**: running reaction i at flux v_i requires an amount of enzyme $v_i/k_{\mathrm{cat},i}$, where $k_{\mathrm{cat},i}$ is the turnover number, and the total enzyme mass is capped by a shared **proteome pool**, so the cell must allocate a finite protein budget across reactions (Equation 2, Equation 6). *Departure.* For *E.*

coli the base is iML1515[18] as the pre-built GECKO ecModel eciML1515[13,14], carrying reference turnovers for 2,560 enzymes; for *M. maripaludis* we build the sMOMENT enzyme layer here on the plain iMR539 reconstruction[19], mapping 527 enzymatic reactions to UniProt masses. A **curation** step differs likewise: *E. coli* required closing four reactions that consume O at negligible enzyme cost — two quinol/menaquinol monooxygenases, a malate oxidase and a cuprous oxidase — which otherwise form a futile O cycle (75% of gross O turnover at the rich-medium optimum) that scrambles the respiratory read-out; closing them routes O through the enzyme-costed terminal oxidase at 0.3% growth cost (catalase and superoxide dismutase, whose high turnover makes their low cost correct, are retained). *M. maripaludis* required a carbon-honest autotroph curation instead: folding the dispensable archaellum glycoprotein out of biomass, repairing a selenocysteinyl-tRNA recycling gap, closing organic-carbon leaks and freeing the pinned methane exchange, giving growth with 99.98% of biomass carbon from CO . *Synechocystis* 6803 starts from the iSynCJ816 reconstruction[20] and, like *E. coli*, uses a **pre-built** enzyme-constrained ecModel — iSynCJ816_STAR[21], an AUTOPACMEN sMOMENT layer (the same total-protein-pool form built by hand for the methanogen) — so no enzyme layer is built here. Its curation is the choice of operating **regime**: we set a light-**saturated** photoautotrophic medium (photon supply non-limiting, CO /bicarbonate available, organic carbon closed) so that carbon fixation, not the temperature-insensitive light reactions, sets the rate — confirmed by a zero photon shadow price at every temperature. No futile-cycle closure was needed (unlike *E. coli*), and net RuBisCO oxygenase (photorespiration) flux is zero at the optimum under the CO -replete conditions, so the growth E_a is Calvin-kinetics-dominated.

Layer 2 — a temperature-dependent enzyme envelope. Each enzyme's turnover rises with temperature following macromolecular rate theory (MMRT)[15] — an Arrhenius law whose activation energy itself changes with temperature (a heat-capacity term), giving a curved, single-peaked response — and above a threshold the enzyme unfolds, modelled by a two-state native unfolded equilibrium (after[4]), the **native fraction** $f_{N,i}(T)$, which collapses around the enzyme's melting temperature T_m. The effective per-flux cost combines both effects (Equation 5), rising both when turnover is slow (cold) and when the enzyme denatures (hot); keying the collapse on T_m ties the upper thermal limit to protein *stability*, distinct from the rising-limb activation energy. *Departure.* The provenance of the per-enzyme thermal parameters differs by organism: for *E. coli* the melting temperatures come from the measured melting proteome[22] and the optima from a sequence-based predictor[23] (~90% grounded, rest at dataset means), refined by DLTKcat[24]; for *M. maripaludis* the energy-core turnovers are measured specific activities[25] converted to k_{cat}, but the optima and melting temperatures are a documented mesophile prior (the sequence/meltome predictors were not run for the archaeon). For *Synechocystis* the turnovers are the ecModel's own (BRENDA/SABIO-RK) values — including the well-measured RuBisCO/Calvin kinetics — while the optima and melting temperatures use the same documented mesophile prior as the methanogen (6803 is a mesophile absent from the sequence/meltome training sets); the curvature is again the single literature MMRT prior.

Layer 3 — proteome allocation. The enzyme budget is split into three broad measured **proteome sectors**[12,16] — metabolic (enzymes carrying flux), biosynthesis (dominated by ribosomes, with a translation cap on growth), and maintenance (housekeeping and stress) — with a temperature-dependent non-growth maintenance NGAM(T) that lifts the hot-side ATP burden and rounds the optimum, and an optional growth law that couples the ribosomal fraction to growth rate (Equation 6–Equation 8). *Departure — the scientifically load-bearing one.* The allocation **law** differs between the taxa and is one of the two terms that explain the activation-energy difference: *E. coli* follows the bacterial Scott growth law, in which the ribosomal fraction *rises* with growth rate (slope $s \approx 0.33$), shrinking the metabolic sector at higher temperature and buffering the activation energy down; *M. maripaludis* instead maintains a **constant** ribosomal fraction and full methanogenesis capacity across growth rate[10] — a documented alternative allocation strategy — so its slope is flat ($s = 0$) and the buffer vanishes. *Synechocystis* follows a third, intermediate law: its ribosomal (RIB) proteome fraction *rises* with growth rate[11], but only weakly in absolute terms over its narrow growth-rate range ($\mu_{\max} \approx 0.1$ h ¹, ~10–20× smaller than *E. coli*'s), so the allocation buffer is small (−0.02 eV). Maintenance is grounded per organism too: an *E. coli* NGAM amplitude for the bacterium, the **measured** *M. maripaludis* maintenance energy[26] for the methanogen, and the **measured** photon-maintenance coefficient of[27] for the phototroph (with the cyanobacterial heat-shock proteome response[28] informing the hot-side maintenance sector).

Layer 4 — an emergent a-priori prediction, with calibration kept separate. For all three organisms the design stance has two distinct parts. *The emergent model is the prediction*: every input is grounded in independent data (the per-enzyme thermal parameters of Layer 2, the measured sector allocation of Layer 3, and the literature magnitude scalars P^{lit} and σ), and nothing is fit to the growth curve, so both the rate magnitude and the whole-organism activation energy **emerge and can be tested** against an independent benchmark. *Calibration is a separate, labelled step*: a Bayesian inverse analysis (Equation 11; the global correction scalars are tabulated in the Supplement) then asks what corrections the measured curve *demands* of the few borrowed or uncertain global scalars — it does not re-fit the grounded per-enzyme parameters — and the emergent prediction and the tuned posterior are kept distinct, so the a-priori gap is quantified rather than papered over.

4.2 Inputs and data provenance: what the model uses for each enzyme

Each model is **per-enzyme**: every modelled enzyme (2,560 for *E. coli*; 527 for *M. maripaludis*; ~1,066 flux-costed reactions for *Synechocystis*) carries its own inputs, applied individually rather than as one global response shared across the proteome. For each enzyme the model uses four things:

- **Flux** — *not an input but a predicted output* of the flux-balance optimisation.
- **Turnover** $k_{\mathbf{cat}}(T)$ — a reference turnover shaped by an MMRT temperature response. For *E. coli* the reference is the GECKO ecModel value[13,14], refined by DLTKcat[24] where a fit exists; for *M. maripaludis* the reference is set by source priority — measured specific activities for the energy-conserving core[25], DLTKcat for the biosynthetic bulk (corrected for a systematic archaeal underprediction with cited literature/BRENDA overrides for the highest-cost offenders and a 1 s^{1} floor), and a dataset-mean fallback.
- **Thermal optimum** $T_{\mathbf{opt}}$ **and curvature** $\Delta C_p^{\ddagger}$ — the optimum is sequence/ML-predicted per enzyme for *E. coli*[23] and a mesophile prior for *M. maripaludis*; the curvature is, for all three, a single independent literature MMRT prior (-4 kJ mol^{1} K^{1}[15]), deliberately not fitted, so the activation energy emerges.
- **Melting temperature** T_m (setting the native fraction) — from the measured *E. coli* melting proteome[22] for the bacterium and a mesophile prior for the archaeon.

Only the pool budget scalars (P^{lit}, σ) and the sector fractions are whole-cell quantities; everything else is per enzyme. Per-enzyme grounding coverage is high but honest: for *E. coli* ~90% of enzymes have data-grounded optima and melting temperatures (the rest at dataset means); for *M. maripaludis* every enzymatic reaction maps to a UniProt mass, the energy core (11 reactions) carries measured kinetics, 293 reactions carry DLTKcat predictions and 223 a mean fallback, while the optima/melting temperatures are the mesophile prior throughout — a stated limitation refined by the calibration's envelope knobs.

4.3 Enzyme abundance and the compensation question

For all three models, per-enzyme **abundances are not measured or set individually**: each enzyme's proteome-mass usage is flux/$k_{\mathrm{cat}}(T) \times \mathrm{MW}$, drawn from the shared proteome pool that the optimiser allocates across all enzymes, with measured fractions supplied only at the coarse sector level. Per-enzyme abundance is therefore an **optimisation outcome, not an input** — a structural property of the GECKO/sMOMENT pool common to all three taxa. This has a consequence worth stating plainly: as temperature rises and an enzyme's k_{cat} falls, or the enzyme begins to unfold, carrying the same flux costs more enzyme mass, so the optimiser draws more of the pool toward those steps — an implicit, mass-balance "compensation" — until the pool binds and growth is throttled. But this is **least-cost allocation under a fixed budget, not regulatory sensing**: no mechanism detects a falling product and up-regulates a gene. The models therefore capture the *demand* side of thermal compensation (more enzyme needed when catalysis is poor) but not active feedback regulation — a genuine limitation of all three.

4.4 Governing equations

Enzyme-constrained flux balance. Growth is the biomass flux v_{bio} found by linear programming under mass balance and enzyme capacity. At steady state

$$\mathbf{S}\,\mathbf{v} = \mathbf{0}, \tag{1}$$

where $\mathbf{S}$ is the stoichiometric matrix and $\mathbf{v}$ the flux vector (mmol gDW^{-1} h^{-1}) with lower/upper bounds encoding reversibility and medium availability. Each reaction is limited by the amount of catalytically competent (native) enzyme,

$$0 \le v_i \le k_{\mathrm{cat},i}(T)\, f_{N,i}(T)\, [E]_i, \tag{2}$$

with turnover $k_{\mathrm{cat},i}(T)$ (Equation 3), total enzyme $[E]_i$, and native fraction $f_{N,i}(T) \in [0,1]$ (Equation 4) — only folded protein catalyses, while denatured protein still occupies the budget. Rather than an enzyme variable per reaction, we impose the equivalent **sMOMENT** total-protein-pool form: a shared, sector-partitioned budget bounds the summed per-flux enzyme demand (Equation 6).

Temperature-dependent turnover. Each enzyme's turnover follows transition-state (Eyring) theory with a macromolecular-rate-theory heat-capacity term[15],

$$k_{\mathrm{cat},i}(T) = \frac{k_B T}{h} \exp\Big(-\frac{\Delta G_i^{\ddagger}(T)}{RT}\Big),\ \Delta G_i^{\ddagger}(T) = \Delta H_i^{\ddagger} + \Delta C_{p,i}^{\ddagger}(T - T_0) - T(\Delta S_i^{\ddagger} + \Delta C_{p,i}^{\ddagger} \ln(T/T_0)), \tag{3}$$

with Boltzmann and Planck constants k_B, h, reference T_0, and per-enzyme activation enthalpy, entropy and heat capacity $\Delta H_i^{\ddagger}, \Delta S_i^{\ddagger}, \Delta C_{p,i}^{\ddagger}$. A negative $\Delta C_p^{\ddagger}$ makes $\ln k_{\mathrm{cat}}$ single-peaked, so the **rising-limb activation energy emerges from Equation 3** rather than being imposed. Turnover is carried relative to its value at each enzyme's optimum $T_{\mathrm{opt},i}$, $\hat{k}_i(T) = k_{\mathrm{cat},i}(T)/k_{\mathrm{cat},i}(T_{\mathrm{opt},i})$, with $\Delta H_i^{\ddagger}$ fixed by the peak condition $\mathrm{d}\ln k_{\mathrm{cat}}/\mathrm{d}T = 0$ at $T_{\mathrm{opt},i}$.

Two-state unfolding and the effective cost. The high-temperature collapse is set by denaturation: a native↔unfolded equilibrium keyed on the melting temperature $T_{m,i}$,

$$f_{N,i}(T) = \frac{1}{1 + \exp(-\Delta G_{u,i}(T)/RT)},\ \Delta G_{u,i}(T) = \Delta H_{TH,i} + \Delta C_{p,u,i}(T - T_H) - T\Delta S_{TS,i} - T\Delta C_{p,u,i} \ln(T/T_S), \tag{4}$$

with universal convergence temperatures $T_H = 373.5$ K, $T_S = 385$ K, and unfolding parameters solved from $T_{m,i}$ (with T_{90} or protein length) via $\Delta G_{u,i}(T_{m,i}) = 0$ and $\Delta G_{u,i}(T_{90}) = -RT_{90} \ln 9$. Combining anchored turnover and native fraction, the per-flux enzyme cost is

$$c_i(T) = \frac{\mathrm{base}_i}{\hat{k}_i(T)\, f_{N,i}(T)\, \lambda_k}, \qquad \mathrm{base}_i = \frac{\mathrm{MW}_i}{k_{\mathrm{cat},i}(T_{\mathrm{opt},i}) \cdot 3600}, \tag{5}$$

with enzyme mass MW_i (kDa = g mmol^{-1}), the factor 3600 converting s^{-1} to h^{-1}, and a global turnover-level multiplier λ_k (`kcat_scale`, nominal 1). Denatured enzyme still occupies the budget but does not catalyse, so cost is inflated by $1/f_{N,i}$ — the origin of the falling limb, keyed on stability T_m and thus decoupled from the rising-limb E_a.

Proteome sectors, the pool budget, and the growth law. The total modelled protein P_{tot} is split into coarse sectors[12,16] summing to one — metabolic f_{metab}, biosynthesis/ribosomal f_{bio}, and maintenance f_{maint} — and growth maximises v_{bio} under a metabolic-pool bound and a biosynthesis (translation) cap,

$$\sum_i c_i(T)\,(v_i^+ + v_i^-) \le f_{\mathrm{metab}}\, P_{\mathrm{tot}}, \qquad \kappa\, v_{\mathrm{bio}} \le f_{\mathrm{bio}}\, P_{\mathrm{tot}}, \tag{6}$$

where $v_i^{\pm} \ge 0$ are forward/reverse fluxes and κ (`translation_coeff`) is the ribosomal demand per unit biomass flux, auto-calibrated once at build so the two caps co-limit at the nominal split (enabling sectors does not move nominal growth). The metabolic budget is grounded independently,

$$P_{\mathrm{metab}} = f_{\mathrm{metab}}\, P_{\mathrm{tot}}, \qquad P_{\mathrm{tot}} = \sigma\, P^{\mathrm{lit}}, \tag{7}$$

with in-vivo saturation σ and literature total protein P^{lit}. Optionally the sectors follow a proteome-conserving growth law in which the ribosomal and metabolic fractions move with growth rate at a common slope s, $f_{\mathrm{bio}}(\mu) = f_{\mathrm{bio},0} + s\mu$ and $f_{\mathrm{metab}}(\mu) = f_{\mathrm{metab},0} - s\mu$ ($\mu = v_{\mathrm{bio}}$), keeping both caps linear,

$$(\kappa - sP_{\mathrm{tot}})\, v_{\mathrm{bio}} \le f_{\mathrm{bio},0} P_{\mathrm{tot}}, \qquad \sum_i c_i(T)(v_i^+ + v_i^-) + sP_{\mathrm{tot}}\, v_{\mathrm{bio}} \le f_{\mathrm{metab},0} P_{\mathrm{tot}}. \tag{8}$$

The slope s is the scientifically load-bearing organism-specific quantity: it is set from each organism's own proteome data (below), and its value is the origin of the allocation term in the E_a decomposition.

Maintenance. A temperature-dependent non-growth maintenance ATP demand rises with temperature[29],

$$\mathrm{NGAM}(T) = a_\mathrm{m}\Big(1 - b\exp\big[\tfrac{-E_m}{k_B}\big(\tfrac{1}{T_\mathrm{ref}} - \tfrac{1}{T}\big)\big]\Big), \quad \text{floored at } T_\mathrm{ref}, \tag{9}$$

with amplitude a_m, $b \approx 0.62$, apparent activation energy $E_m \approx 0.5$ eV, $T_\mathrm{ref} = 298.15$ K and $k_B = 8.617 \times 10^{-5}$ eV K ^{1}, imposed as $v_\mathrm{ATPM} \geq \mathrm{NGAM}(T)\, f_\mathrm{maint}/f_\mathrm{maint}^\mathrm{nom}$; the growth-associated maintenance (GAM) is the ATP demand carried in the biomass reaction. At each grid temperature FBA maximises $\mu(T) = \max_\mathbf{v} v_\mathrm{bio}$ subject to Equation 1, Equation 6 and the maintenance floor, tracing the organismal TPC.

The activation energy. From $\mu(T)$ we read standard descriptors — the optimum $T_\mathrm{opt} = \arg\max_T \mu$, the maximum rate r_max, the critical limits $CT_\mathrm{min}/CT_\mathrm{max}$ (where μ falls to 5% of r_max), and the niche width. The naive Boltzmann–Arrhenius activation energy — the slope of $\ln\mu$ against $-1/(k_B T)$ over a rising-limb window — is window-dependent (Results, Section 2.2), so we report instead the **window-independent Sharpe–Schoolfield** E[8], the rising-limb activation energy of the high-temperature-inactivation model

$$\mu(T) = r_{T_\mathrm{ref}}\,\frac{\exp[(E/k_B)(1/T_\mathrm{ref} - 1/T)]}{1 + \exp[(E_h/k_B)(1/T_h - 1/T)]}, \tag{10}$$

fit to the whole calibrated curve by nonlinear least squares at a fixed $T_\mathrm{ref} = 20$ °C, where E is the reported E_a and E_h, T_h describe the high-T deactivation. The same fit is applied identically to model and observed curves and to all three organisms.

Bayesian calibration. Each model is calibrated to one measured growth TPC with a gradient-free ensemble sampler (emcee), warm-started at the posterior mode and run to an autocorrelation-based stop. The deterministic simulator $\theta \mapsto \mu(T)$ is combined with a Gaussian discrepancy likelihood

$$\ln L(\theta) = -\tfrac{1}{2}\sum_t \Big[\frac{(\mu_t^\mathrm{obs} - \mu(T_t;\theta))^2}{\mathrm{sd}_t^2 + \sigma_\mathrm{disc}^2} + \ln\big(2\pi(\mathrm{sd}_t^2 + \sigma_\mathrm{disc}^2)\big)\Big], \tag{11}$$

with per-point measurement SD sd_t (zero where none is reported) and a free model-discrepancy scale σ_disc. Only a small set of **global scalars** is freed — $\{\lambda_k$ (`kcat_scale`), σ, dT_opt, `topt_scale`, `dCp_scale`, dT_m, `tm_scale`, f_metab, f_maint, `ngam_scale`, `ngam_steepness`, $\sigma_\mathrm{disc}\}$ — each transforming the *whole* grounded per-enzyme distribution while preserving its shape (a uniform shift, stretch or rescale), never a per-enzyme free parameter. Positive-only parameters are sampled in log space; priors are provenance-based (measured sector fractions tight; the in-vivo saturation a broad Normal on 0.45; magnitude and envelope levers LogNormal about their nominal). The emergent (nothing-fit) model and the calibrated posterior are kept distinct throughout.

The control-weighted activation-energy decomposition. To ask what sets each E_a we decompose the Sharpe–Schoolfield E, by metabolic control analysis, into a control-weighted mean of the per-enzyme rising-limb activation energies and four named couplings relative to the naive (unweighted) enzyme mean:

$$E_a^\mathrm{org} = \overline{E_i} + \underbrace{\big(\textstyle\sum_i \hat{C}_i E_i - \overline{E_i}\big)}_{\text{control weighting}} + \Delta_\mathrm{alloc} + \Delta_\mathrm{maint} + \Delta_\mathrm{agg}. \tag{12}$$

The construction is non-circular: the per-enzyme E_i — the rising-limb Arrhenius activation energy of each enzyme's turnover $k_{\mathrm{cat},i}(T)$ alone (the folding term is the high-T deactivation, contributing a negligible -0.008 eV control-weighted) — are model *inputs*, and the result is how the organism value aggregates and departs from them. The growth flux-control coefficients $\hat{C}_i = \partial\ln\mu/\partial\ln k_{\mathrm{cat},i}$ are computed by symmetric perturbation and re-solve and normalised over the controlling set. The **allocation** term is isolated by disabling the growth-law coupling (Equation 8), the **maintenance** term by flattening $\mathrm{NGAM}(T)$, and the **aggregation** term is computed directly as the Sharpe–Schoolfield E of the kinetic-backbone aggregate curve minus the control-weighted per-enzyme mean. All three organisms are analysed on the calibrated model at their operating point, each carrying its own sector/growth-law layer (and, for the phototroph, under the light-saturated medium with the photon supply non-binding). For Figure 1 the saved calibration draws were propagated through Equation 10 to obtain the E_a posteriors.

4.5 Model hierarchy: how the components link

The model is a hierarchy that turns grounded per-enzyme thermal parameters into an organismal curve, each level feeding only the next, identically for all three taxa. **(i)** Every enzyme carries three grounded thermal parameters — its optimum $T_{\mathrm{opt},i}$, its melting temperature $T_{m,i}$ and its turnover curvature $\Delta C^{\ddagger}_{p,i}$. **(ii)** These set the two per-enzyme temperature responses: the optimum-anchored turnover $\hat{k}_i(T)$ (Equation 3) and the native fraction $f_{N,i}(T)$ (Equation 4). **(iii)** The two combine into an effective per-flux enzyme cost $c_i(T)$ (Equation 5). **(iv)** The costs enter the sector-partitioned proteome constraints — metabolic pool, biosynthesis cap and temperature-dependent maintenance (Equation 6, Equation 9) — optionally coupled through the growth law (Equation 8). **(v)** FBA maximises biomass flux at each temperature, tracing the TPC $\mu(T)$. **(vi)** The curve is reduced to descriptors and the Sharpe–Schoolfield activation energy (Equation 10). **(vii)** For calibration and the decomposition, the global correction knobs (Supplement) deform the whole per-enzyme distribution at level (i). The forward chain is thus *grounded parameters → per-enzyme responses → cost → sector constraints + growth law + NGAM(T) → FBA growth → descriptors*, with no growth-curve information entering anywhere in the emergent model.

4.6 Relationship to the Li et al. 2021 yeast etc-GEM

Both models are direct descendants of the yeast etc-GEM of[4], and we **follow it for the thermal core**: the transition-state turnover $k_{\mathrm{cat}}(T)$ (Equation 3), the two-state native-fraction denaturation keyed on T_m (Equation 4), a temperature-dependent maintenance term, and the enzyme-constrained proteome-pool idea. We **depart from and upgrade** it in five deliberate ways. First, **organism**: rather than *S. cerevisiae*, we build a bacterium, an archaeon and a cyanobacterium — to our knowledge the first etc-GEMs of their kind for these taxa — spanning the three energy metabolisms of the[1] ordering, with the archaeon requiring the sMOMENT enzyme layer built here (the bacterium and cyanobacterium use pre-built enzyme-constrained models). Second, **grounded not fitted per-enzyme parameters**: we fix each enzyme's T_{opt}, T_m and curvature from independent data, so the TPC is an a-priori forward prediction, whereas Li et al. infer the per-enzyme thermal parameters by fitting the growth data — the opposite epistemic stance, and grounding is what lets us *test* the emergent activation energy against an independent benchmark. Third, a **sector-partitioned proteome** with an explicit translation cap, replacing the single undivided pool, which makes allocation an attributable control on the rate. Fourth, an **organism-specific allocation law** — the Scott rising-ribosome law for *E. coli*, the Müller constant-ribosome strategy for *M. maripaludis*, and the Jahn shallow-ribosome law for *Synechocystis* — rather than a single fixed coupling. Fifth, **inference by likelihood, not ABC**: we calibrate a low-dimensional set of global correction knobs by likelihood-based ensemble MCMC (emcee), kept as a separate inverse layer on top of the emergent prior, exploiting the fact that the etc-GEM is a deterministic simulator with a Gaussian residual model; keeping the emergent prediction and the tuned posterior separate is what lets us report an honest a-priori gap. For quantifying the activation energy we further add the window-independent Sharpe–Schoolfield E (Equation 10) and the control-weighted E_a decomposition (Equation 12), neither of which appears in the yeast model.

4.7 Validation

For each organism the emergent (nothing-fit) model is first tested a-priori against an independent exact-strain growth TPC, and only then calibrated. *E. coli* is validated against the K-12 MG1655 rich-medium (BHI) curve of[6] over 7–46 °C; *M. maripaludis* against the type-strain H /CO curve of[5] over 15–52 °C; and *Synechocystis* 6803 against the light-saturated photoautotrophic growth curve of[7] over 23–38 °C, with the independent photosynthesis-flux (O -evolution) TPC of[9] as a second, non-fitted cross-check. The reported activation energy is the window-independent Sharpe–Schoolfield E of Equation 10, applied identically to model and observed curves. It agrees for all three: the *E. coli* model gives $E_a = 0.68$ eV against an observed 0.56 (on the ~0.65 eV benchmark), *M. maripaludis* gives 1.06 against 1.04, and *Synechocystis* gives 0.57, whose carbon-fixation-flux E (0.56) matches the independent Inoue photosynthesis-flux value (0.52); the three calibration posteriors reproduce the full ordering (Results, Section 2.1, Section 2.2). Because the per-enzyme activation energies are model inputs, this agreement on each organism-level value is a genuine prediction rather than a recovered fit.

4.8 Taxon-specific construction and departures

The three models instantiate the same framework but differ, layer by layer, in ways that are either technical (base reconstruction, parameter provenance) or scientifically load-bearing (the allocation law and the controlling pathway). Table 4 summarises the contrasts; the most consequential are the kcat provenance for the energy core, the operating regime for the phototroph, and — the term that carries part of the E_a ordering — the allocation law.

Base network and enzyme layer. *E. coli* starts from iML1515[18] as a pre-built GECKO ecModel (eciML1515)[13,14] carrying reference turnovers and masses for 2,560 enzymes; *M. maripaludis* starts from the plain iMR539 reconstruction[19], onto which we build the sMOMENT enzyme layer here (527 enzymatic reactions mapped to UniProt masses); *Synechocystis* starts from iSynCJ816[20] with the pre-built AUTOPACMEN sMOMENT ecModel iSynCJ816_STAR[21] (~1,066 flux-costed reactions), the cyanobacterial analog of the pre-built *E. coli* ecModel.

Curation. *E. coli* required closing four uncosted O -sink reactions (two quinol monooxygenases, a malate:O oxidoreductase, a cuprous oxidase) that form a futile O cycle. *M. maripaludis* required a carbon-honest autotroph curation: folding the dispensable archaellum glycoprotein out of biomass, repairing a selenocysteinyl-tRNA recycling gap, closing organic- carbon leaks and freeing the pinned methane exchange, and supplying two non-synthesisable vitamin cofactors as trace supplements. *Synechocystis* required no futile-cycle closure; its "curation" is the choice of the **light-saturated** operating regime (photon non-limiting, organic carbon closed), which keeps the temperature-insensitive light reactions non-binding so carbon-fixation kinetics set the rate (verified by a zero photon shadow price across the TPC).

Turnover provenance. *E. coli* uses the GECKO built-in k_{cat} refined by DLTKcat[24]. For *M. maripaludis* the energy-conserving core was set from measured specific activities (the electron-bifurcation hydrogenase/heterodisulfide-reductase complexes[25], converted to k_{cat} via complex masses), the biosynthetic bulk from DLTKcat corrected for a systematic archaeal underprediction (28% of predictions below 1 s ^{1}) with cited literature/BRENDA overrides for the highest-cost offenders and a documented 1 s ^{1} floor, and a mean fallback; Mcr — the E_a target — was sourced explicitly ($k_{\text{cat}} \approx 59$ s ^{1}, 3–294 s ^{1} range, swept in Section 2.4). *Synechocystis* uses the ecModel's own BRENDA/SABIO-RK turnovers, including the well-measured RuBisCO/Calvin-cycle kinetics; the carbon-fixation turnover is swept for robustness (Section 2.4, Supplement) as the phototroph analog of the Mcr sweep.

Optima and stability. *E. coli* $T_{\text{opt},i}$ and $T_{m,i}$ are grounded per enzyme (sequence-based optimum predictor[23]; measured melting proteome[22]) for ~90% of enzymes. For *M. maripaludis* and *Synechocystis* these predictors were not run (both are mesophiles absent from the training sets); both use a documented mesophile prior (a flagged limitation refined by the calibration's dT_{opt}/dT_m envelope knobs).

The allocation law (load-bearing departure). Each organism uses its **own measured** law. *E. coli* uses the bacterial Scott growth law[12]: the ribosomal fraction *rises* with growth rate (slope $s \approx 0.33$), shrinking the metabolic sector at higher temperature and buffering E_a downward (−0.13 eV). *M. maripaludis* holds a **constant** ribosomal fraction across growth rate[10] ($s = 0$, a flat law), so the buffer vanishes (+0.00 eV). *Synechocystis* rises weakly: its RIB sector increases with growth rate[11] but over a ~10–20× smaller growth-rate range, so the absolute reallocation — and the buffer — is small (−0.02 eV). These physiological differences, not modelling choices, are among the terms that set the E_a ordering (Results, Section 2.3).

Maintenance, medium and saturation. Maintenance NGAM(T) uses an *E. coli* amplitude for the bacterium, the **measured** *M. maripaludis* maintenance energy[26] for the methanogen, and the **measured** *Synechocystis* photon-maintenance coefficient[27] for the phototroph (its heat-shock proteome response[28] informing the hot-side sector). Validation/calibration is against BHI-rich growth[6] for *E. coli*, H /CO growth[5] for the methanogen, and light-saturated autotrophic growth[7] (with the Inoue flux TPC[9] as a cross-check) for the phototroph. All take the in-vivo saturation σ from the same literature range[30,31]. Full per-model construction, parameter and provenance tables, and robustness analyses are in the Supplement.

Table 4: Where the three etc-GEMs depart, by model layer. The allocation-law and controlling-backbone rows are the scientifically load-bearing differences: each organism's activation energy is set by *where* its growth control sits (high- vs low-E_a backbone) and its *own measured* allocation strategy. Saturation σ is 0.45 (0.4–0.5)[30,31] for all three.

Layer	*M. maripaludis* (methanogenesis)	*E. coli* (respiration)	*Synechocystis* 6803 (photosynthesis)
Base GEM	iMR539[19] → our sMOMENT	iML1515[18] → GECKO eciML1515	iSynCJ816[20] → sMOMENT ecModel[21]
Curation	carbon-honest autotroph	4 uncosted O sinks closed	light-saturated regime (photon non-limiting)
k_{cat}	measured core[25] + DLTKcat + floor	ecModel built-in + DLTKcat	ecModel BRENDA/SABIO-RK (RuBisCO/Calvin measured)
T_{opt}/T_m	mesophile prior	sequence/meltome-grounded (~90%)	mesophile prior
Allocation law	**Müller: constant ribosome ($s = 0$)**	**Scott: ribosome rises ($s \approx 0.33$)**	**Jahn: weak rise, small range (s small)**
Controlling backbone	methanogenesis (high-E_a)	broad (no single backbone)	Calvin/carbon fixation (low-E_a)
Maintenance	measured NGAM[26]	*E. coli* NGAM(T)	measured photon-maint.[27]
Medium / curve	H /CO , Jones 1983	BHI rich, Van Derlinden 2012	light-sat. autotrophy, Zavrel 2015 (+ Inoue 2001)

5 Data and code availability

All figures and tables are generated from the analysis outputs of the three etc-GEMs and their cross-organism comparison. The complete model construction for all three organisms, the parameter and data-source tables, the window-sensitivity audit, the calibration diagnostics, and the robustness analyses are provided in the Supplement.

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